

Exponential Excess-Rank Tails and the Half-Logarithmic Postblock Clock

From Endpoint Gaussian Resolution to a Concrete Stage-A Factorization Gate

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Abstract

RH-77 found that the physical postblock states are extraordinarily compressible, while RH-81 identified an all-level effective-rank theorem as the preferred remaining Stage-A gate. This paper replaces that qualitative request by a specific factorization problem.

RH-16 constructed the endpoint-projected Gaussian resolution operator $\mathcal{R}_\sigma$ and proved its half-logarithmic threshold-rank law. We sharpen that result from a count to a full approximation-number tail. Let $J_\sigma = \max\{k : \delta_k \geq \sigma\}$ for the geometric endpoint ladder. If $\delta_{k+1}/\delta_k \leq q < 1$ eventually and the unresolved projected-row energy satisfies $F(t) \leq Ct^2$, then for every $\ell \geq 0$,

$$\tau_{J_\sigma+\ell}(\mathcal{R}_\sigma) \leq \sqrt{\frac{C}{1-q^2}} q^\ell,$$

where τ_r is the optimal Hilbert–Schmidt rank- r residual. Since $J_\sigma = \log(1/\sigma)/(2\log \lambda) + O(1)$, every extra rank beyond the half-logarithmic clock buys exponential tail decay.

We then prove an ideal-property transfer theorem. If a physical postblock state factors as

$$B_\sigma = U_\sigma \mathcal{R}_\sigma V_\sigma + E_\sigma,$$

then its rank tail is bounded by the endpoint tail multiplied by $\|U_\sigma\| \|V_\sigma\|$, plus $\|E_\sigma\|_2$. Polylogarithmic outer, remainder, and observability bounds therefore imply the RH-78 effective-rank corridor with rank $O(\log(1/\sigma))$.

A 192-bit Arb audit tests the canonical schedule $r_\sigma = \lceil \log(1/\sigma)/(2\log \lambda) \rceil + 2$ on all ten archived postblock channels. The ranks are four through seven, the maximum relative residual is 2.34×10^{-7} , and the maximum full-future Hardy perturbation is 5.05×10^{-9} . In the RH-16 linear endpoint-row model, the same clock-plus-two tail remains below 9.19×10^{-8} from $\sigma = 10^{-2}$ to 10^{-12} .

The actual endpoint-to-postblock factorization is not proved here. It is now the explicit first-open Stage-A gate. Unconditional Stage A4, Hilbert–Polya, and the Riemann Hypothesis remain open.

Keywords: approximation numbers; effective rank; Gaussian resolution; postblock state; Hilbert–Schmidt tail; small noise.

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1 From a rank observation to a factorization gate

The RH-77 production audit showed that rank two captures at least 99% and rank four at least 99.9999% of every archived postblock state [2]. RH-78 proved that a polylogarithmic rank and future residual are enough for Stage A1 [4]. What remained unclear was why the physical family should possess such a rank law at all dyadic levels.

There is already an analytic logarithmic-rank object in the program. RH-16 associated normalized Gaussian rows to the endpoint ladder

$$\delta_k = C_b \lambda^{-2k} (1 + o(1)), \quad \lambda = 1.678573510428322 \dots, \quad (1)$$

removed their common endpoint vector, and obtained a Hilbert–Schmidt resolution operator $\mathcal{R}_\sigma$ [3]. Its squared Hilbert–Schmidt norm and fixed-threshold rank are

$$H_\sigma + O(1), \quad H_\sigma := \frac{\log(1/\sigma)}{2 \log \lambda}. \quad (2)$$

A threshold count alone does not control the Frobenius tail required by RH-77. The first result supplies that missing estimate.

2 Exponential tail beyond the half-log clock

Let ψ_{σ, δ_k} be the normalized endpoint Gaussian row, Q_σ the orthogonal complement of the limiting endpoint row, and

$$\mathcal{R}_\sigma e_k = Q_\sigma \psi_{\sigma, \delta_k}. \quad (3)$$

Write

$$F(t) = \|Q_\sigma \psi_{\sigma, \delta}\|^2, \quad t = \delta/\sigma. \quad (4)$$

The exact Gaussian geometry of RH-16 gives $F(t) \leq Ct^2$ for $0 \leq t \leq 1$.

For a Hilbert–Schmidt operator T , define

$$\tau_r(T) = \inf_{\text{rank } L \leq r} \|T - L\|_2 = \left(\sum_{j>r} s_j(T)^2 \right)^{1/2}. \quad (5)$$

Theorem 2.1 (Exponential excess-rank tail). *Assume that for all $k \geq k_0$,*

$$0 < \frac{\delta_{k+1}}{\delta_k} \leq q < 1. \quad (6)$$

Let $J_\sigma = \max\{k : \delta_k \geq \sigma\}$. For all sufficiently small σ and every integer $\ell \geq 0$,

$$\boxed{\tau_{J_\sigma + \ell}(\mathcal{R}_\sigma) \leq \sqrt{\frac{C}{1 - q^2}} q^\ell.} \quad (7)$$

Moreover $J_\sigma = H_\sigma + O(1)$.

Proof. Truncate (3) after column $J_\sigma + \ell$. The truncation has rank at most $J_\sigma + \ell$, hence

$$\tau_{J_\sigma + \ell}(\mathcal{R}_\sigma)^2 \leq \sum_{m \geq 1} F\left(\frac{\delta_{J_\sigma + \ell + m}}{\sigma}\right).$$

Maximality gives $\delta_{J_\sigma + 1} < \sigma$. Iterating (6) therefore gives

$$\frac{\delta_{J_\sigma + \ell + m}}{\sigma} \leq q^{\ell + m - 1} \leq 1.$$

Using $F(t) \leq Ct^2$ and summing the geometric series proves (7). The final clock identity follows from (1), as in RH-16. $\square$

Corollary 2.2 (Logarithmic accuracy rank). *For every target $0 < \varepsilon < 1$, there is a rank*

$$r_{\sigma,\varepsilon} = H_\sigma + O\left(1 + \log \frac{1}{\varepsilon}\right) \quad (8)$$

such that

$$\tau_{r_{\sigma,\varepsilon}}(\mathcal{R}_\sigma) \leq \varepsilon.$$

In particular, a polylogarithmic accuracy target costs only $O(\log \log(1/\sigma))$ ranks beyond the half-log clock.

This strengthens the RH-16 threshold theorem without changing its geometric input. The proof uses the unresolved end of the endpoint ladder, where the projected Gaussian rows vanish quadratically.

3 Transfer through a physical postblock factorization

Let B_σ denote the postblock state $A_\sigma^{M_\sigma} X_\sigma$. The desired bridge is a factorization through the endpoint resolution operator.

Theorem 3.1 (Endpoint-to-postblock ideal-property transfer). *Suppose*

$$B_\sigma = U_\sigma \mathcal{R}_\sigma V_\sigma + E_\sigma, \quad (9)$$

where U_σ, V_σ are bounded and E_σ is Hilbert–Schmidt. Then for every $\ell \geq 0$,

$$\tau_{J_\sigma+\ell}(B_\sigma) \leq \|U_\sigma\| \|V_\sigma\| \sqrt{\frac{C}{1-q^2} q^\ell} + \|E_\sigma\|_2. \quad (10)$$

Proof. Let $R_{\sigma,r}$ be the column truncation used in Theorem 2.1. The operator $U_\sigma R_{\sigma,r} V_\sigma$ has rank at most r . The ideal property of the Hilbert–Schmidt class gives

$$\|B_\sigma - U_\sigma R_{\sigma,r} V_\sigma\|_2 \leq \|U_\sigma\| \|V_\sigma\| \|\mathcal{R}_\sigma - R_{\sigma,r}\|_2 + \|E_\sigma\|_2.$$

Insert (7) and minimize over rank- r approximants [1]. □

Corollary 3.2 (Sufficient Stage-A rank corridor). *Assume (9) and that*

$$\|U_\sigma\| \|V_\sigma\|, \quad \|E_\sigma\|_2, \quad \|\mathcal{O}_\sigma\|^{1/2} \leq \text{polylog}(1/\sigma), \quad (11)$$

where $\mathcal{O}_\sigma$ is the future observability Gramian. Then the reduced future and its residual are polylogarithmic at rank $r_\sigma = O(\log(1/\sigma))$. Consequently the RH-78 effective-rank premise holds, provided the already isolated finite-prefix bounds hold.

Proof. The RH-16 Hilbert–Schmidt law gives $\|\mathcal{R}_\sigma\|_2 = O(\sqrt{\log(1/\sigma)})$. Thus the rank- r factorized approximant has polylogarithmic Hilbert–Schmidt norm. Multiplication by $\|\mathcal{O}_\sigma\|^{1/2}$ controls its future Hardy seminorm. The same observability factor applied to (10) controls the residual. RH-78 then supplies the stated composition. □

The corollary is deliberately a criterion. This paper does not assert (9) for the physical folded-Gaussian postblock state.

σ	H_σ	rank	max relative residual	max future error
0.16	1.7691	4	2.340×10^{-7}	5.044×10^{-9}
0.08	2.4383	5	7.528×10^{-11}	2.111×10^{-12}
0.04	3.1074	6	1.937×10^{-11}	7.256×10^{-13}
0.02	3.7765	6	2.870×10^{-10}	6.262×10^{-12}
0.01	4.4457	7	1.048×10^{-10}	1.752×10^{-12}

Table 1: Outward-rounded maxima over the two physical channels.

4 Validated clock audit

The frozen audit uses exactly the RH-77 matrices and horizons. Binary64 matrix entries and SVD candidates are lifted exactly into Arb, and all matrix powers and residual norms are propagated at 192-bit precision. The rank is fixed before inspecting each channel:

$$r_\sigma = \lceil H_\sigma \rceil + 2. \quad (12)$$

The coarsest channel is the worst one. Across all ten channels, energy capture is at least 0.999999999999945. The clock-plus-two rule is therefore substantially stronger than the fixed rank-four engineering gate of RH-77.

As an independent model check, the RH-16 linear-row Gram operator was replayed on six noise levels from 10^{-2} to 10^{-12} . The optimal Hilbert–Schmidt tail at (12) lies between 8.93×10^{-8} and 9.19×10^{-8} . This floating Gram diagnostic is consistent with a fixed excess rank, but the theorem comes from Theorem 2.1, not from the fit.

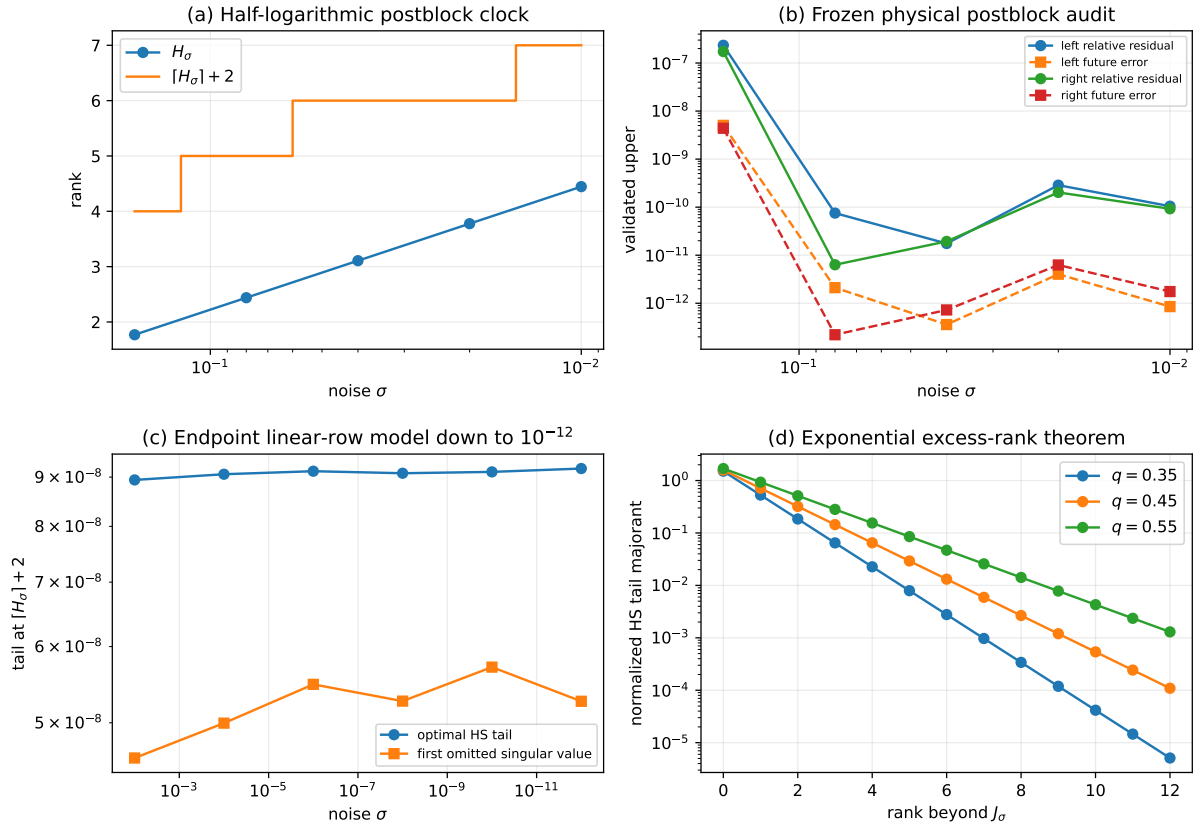


Figure 1: Half-log clock, physical postblock residuals, endpoint-row model tails, and the geometric excess-rank majorant.

5 Route consequence and boundary

The preferred Stage-A route is now narrower. It is unnecessary to guess a new singular-value law for the full physical matrices. The endpoint resolution operator already has the exact logarithmic rank and exponential tail required downstream. The next task is to build (9), with factor norms and remainder compatible with (11).

This paper proves the endpoint excess-rank theorem and the abstract factorization transfer. It validates the half-log plus two schedule only for the five archived frozen families. It does not prove the physical factorization, uniform Stage A1, unconditional Stage A4, an A5 relative determinant, a self-adjoint Hilbert–Polya operator, a $T \log T$ law, a prime-power trace formula, a zeta-zero identification, or the Riemann Hypothesis.

References

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